

**Question 1:**

Determine whether each of the electron configurations given below is an inert gas (Group 0), a halogen (Group VIIA), an alkali metal (Group IA), an alkaline earth metal (Group IIA), or a transition metal (Groups IIB & IIIB). Justify your choices.

- a)  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$
- b)  $1s^2 2s^2 2p^6 3s^2 3p^6$
- c)  $1s^2 2s^2 2p^5$
- d)  $1s^2 2s^2 2p^6 3s^2$
- e)  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$
- f)  $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

**Question 2:**

I) Briefly cite the main differences between ionic, covalent, and metallic bonding.

II) For each of the following compounds, state whether the bonding is essentially metallic, covalent, ionic, or van der Waals. If ionic and covalent bonds are involved in the bonding of any of the compounds listed, calculate the percentage ionic character in the compound.

- a) brass (a copper-zinc alloy)
- b)  $\text{TiO}_2$
- c) Diamond
- d)  $\text{ZnTe}$
- e) Rubber
- f)  $\text{H}_2\text{O}$  within the molecules
- g)  $\text{H}_2\text{O}$  between the molecules
- h) Solid xenon
- i) Bronze
- j) Nylon
- k)  $\text{CaCl}_2$

**Question 3:**

For a  $K^+-Cl^-$  ion pair, attractive and repulsive energies  $E_A$  and  $E_R$ , respectively, depend on the distance between the ions  $r$ , according to  $E_A = -\frac{1.436}{r}$  and  $E_R = \frac{5.8 \times 10^{-6}}{r^9}$ .

For these expressions, energies are expressed in electron volts (eV) per  $K^+-Cl^-$  pair, and  $r$  is the distance in nanometers.

The net energy  $E_N$  is just the sum of the two expressions above.

- a) Calculate the equilibrium interionic spacing ( $r_0$ ).
- b) Calculate the bonding energy ( $E_0$ ).

**Question 4:**

- a) Calculate the force of attraction using the attractive energy - interatomic separation relationship found in your textbook between a  $K^+$  and an  $O^{2-}$  ion, the centers of which are separated by a distance of 1.5 nm.
- b) Calculate the repulsive force if 1.5 nm is the equilibrium interatomic spacing.

**Question 5:**

- a. Calculate the number of atoms in a gold wire that is 0.70 mm in diameter and 8.0 cm. in length. The density of gold is  $19.3 \text{ g/cm}^3$
- b. Calculate the number of atoms in one ton of iron. The atomic weight of iron is  $55.85 \text{ g/mol}$ .